SAEON Product Backlog

Ordered work and taskboard

Reporting period: 28 July to 23 October 2026, Tuesday to Friday. Document revision: 27 October 2026.

The ESP32 WROOM-32 reads the TR-525M rain gauge. DHT11 is used for testing; SHT31 is the temperature and humidity sensor selected for the final product. Firebase is the database platform.

The existing status snapshot runs to 23 October 2026: 43 original task cards are reported Completed and six deployment or field-validation cards remain In Progress. Completion labels require supporting evidence; they do not establish final field acceptance.

PB01–PB13 retain the original sprint groups and task statuses. US references link them to user stories. All are Must items for the agreed scope. Completed means reported complete and awaiting the relevant evidence checks. PB17–PB19 are current follow-up actions, not backdated work. PB14–PB16 remain optional future enhancements.

PB01 Project Initiation & System Architecture

Week 1 | 28 July 2026 to 31 July 2026 | US01, US02, US03, US04, US13

Product Backlog	In Progress	Completed
1.01 Define Project Scope & Board Setup		Completed
1.02 Component Verification & Sourcing		Completed
1.03 Development Environment Configuration		Completed
1.04 Architecture Diagram & Data Flow		Completed

PB02 Core Firmware & Sensor Interfacing

Week 2 | 4 August 2026 to 7 August 2026 | US01, US02, US11

Product Backlog	In Progress	Completed
2.01 ESP32 Base Setup & Serial Output		Completed
2.02 DHT11 Testing and SHT31 Final Sensor Requirements		Completed
2.03 TR-525M Rain Gauge Logic		Completed
2.04 Firmware Formatting & Low Power Mode		Completed

PB03 HiveMQ Cloud Broker Setup

Week 3 | 11 August 2026 to 14 August 2026 | US03, US04, US09

Product Backlog	In Progress	Completed
3.01 Broker Provisioning		Completed
3.02 Topic Structuring & Access Control		Completed
3.03 Network Connection & Publish Testing		Completed
3.04 Firebase Storage Integration		Completed

PB04 Cellular Communication (SIM7000G)

Week 4 | 18 August 2026 to 21 August 2026 | US03, US04

Product Backlog	In Progress	Completed
4.01 SIM Module Wiring & Power		Completed
4.02 Network Bearer Configuration		Completed
4.03 Cellular MQTT to Firebase Integration		Completed
4.04 Stability & Error Handling		Completed

PB05 SAEON Dashboard Base Application

Week 5 | 25 August 2026 to 28 August 2026 | US05, US06, US09

Product Backlog	In Progress	Completed
5.01 App Skeleton & Routing		Completed
5.02 Real-Time Visualization		Completed
5.03 Historical Data & Filtering		Completed
5.04 Cloud Command Prep		Completed

PB06 Power Supply & Solar Integration

Week 6 | 1 September 2026 to 4 September 2026 | US11

Product Backlog	In Progress	Completed
6.01 Battery & Charge Controller Setup		Completed
6.02 Solar Panel Integration		Completed
6.03 Mains Power Alternative Testing		Completed
6.04 System Power Flow Audit		Completed

PB07 Advanced UI & Health Diagnostics

Week 7 | 8 September 2026 to 11 September 2026 | US07, US08, US10

Product Backlog	In Progress	Completed
7.01 Develop Station Health Diagnostics UI Component		Completed
7.02 Implement CSV Export Functionality		Completed
7.03 Mobile & Tablet Responsiveness		Completed
7.04 Full End-to-End Data Pipeline Ping Test		Completed

PB08 Hardware Finalization & PCB Soldering

Week 8 | 15 September 2026 to 18 September 2026 | US12

Product Backlog	In Progress	Completed
8.01 Circuit Transfer & Layout		Completed
8.02 PCB Soldering		Completed
8.03 Enclosure Preparation		Completed
8.04 Component Mounting & Sealing		Completed

PB09 Pre-Deployment Auditing & Calibration

Week 9 | 22 September 2026 to 25 September 2026 | US11, US12

Product Backlog	In Progress	Completed
9.01 Power Drain Audit		Completed
9.02 Solar Charging Verification		Completed
9.03 Full System Deployment	In Progress	
9.04 Baseline Calibration	In Progress	

PB10 48-Hour Field Stress Test

Week 10 | 29 September 2026 to 2 October 2026 | US01, US02, US03, US11, US12

Product Backlog	In Progress	Completed
10.01 Stress Test Initiation	In Progress	
10.02 Active Monitoring	In Progress	
10.03 Manual Field Measurements	In Progress	
10.04 Data Auditing & Conclusion	In Progress	

PB11 Presentation Formatting & Content

Week 11 | 6 October 2026 to 9 October 2026 | US13

Product Backlog	In Progress	Completed
11.01 Presentation Outlining		Completed
11.02 Technical Diagrams & Architecture Slides		Completed
11.03 Dashboard & Software Slides		Completed
11.04 Technical Documentation		Completed

PB12 Rehearsal & Submission Prep

Week 12 | 13 October 2026 to 16 October 2026 | US13

Product Backlog	In Progress	Completed
12.01 Slide Deck Finalization		Completed
12.02 Group Rehearsal 1		Completed
12.03 Timed Rehearsal & Feedback		Completed
12.04 Final Compilation		Completed

PB13 Project Buffer & Final Review

Week 13 | 20 October 2026 to 23 October 2026 | US12, US13

Product Backlog	In Progress	Completed
13.01 Buffer & Prototype Documentation Sign-off		Completed

Current follow-up work

PB17 Verify the final SHT31 configuration

Priority: Must | Status: Product Backlog | US02 | Lead: Redah with Qaasim

Record SHT31 wiring, firmware configuration, temperature and humidity readings, error handling and reference checks. Selection is confirmed; final validation evidence remains to be checked.

PB18 Verify Firebase authentication and access rules

Priority: Must | Status: Product Backlog | US04, US09 | Lead: Qaasim with Angelo

Record the implemented Firebase database service, sign-in checks, restricted access tests and the backend ingestion configuration.

PB19 Complete the assessment evidence package

Priority: Must | Status: Product Backlog | US13 and graduate attributes | Lead: Ethan with Rameez

Confirm individual logbooks, attach evidence, personalise graduate-attribute reflections, verify the assessment period and publish or update the GitHub portfolio.

Optional future enhancements

ID	Item	Priority and status
PB14	Pressure, wind and UV sensing	Could / Product Backlog
PB15	GPS location reporting	Could / Product Backlog
PB16	Over-the-air firmware updates	Could / Product Backlog

Traceability and taskboard walkthrough

Task leads, proposed effort estimates and evidence requirements are recorded in the Sprint Backlog. The Product Backlog is the ordered list of work; a sprint backlog is the selected work and delivery plan for a particular week. Original daily card titles have been clarified to reflect Firebase and the testing/final-sensor distinction.

Taskboard walkthrough

Current review on 27 October 2026. These responses record the present follow-up actions, not meetings backdated to the reporting period.

What do we need to do to advance this piece of work

Confirm the SHT31 final configuration and Firebase evidence, finish installation and baseline calibration, then complete and audit the 48-hour field test.

Is anyone working on anything that is not on the board

Ethan's 4–7 August entries include simulation work, HTTP connection investigation and a supervisor meeting about a pre-seed presentation. Cross-reference these as historical support activities; do not replace them with planned sprint tasks. Record any further unlisted work when confirmed.

What do we need to finish as a team

Close the six open field tasks, confirm PB17–PB19 evidence, verify individual daily entries and complete the graduate-attribute and GitHub portfolio evidence.

Are there any bottlenecks or blockers to the flow of work

Open dependencies include installation, calibration, final-sensor evidence and field-test records. Site access, modem power and connectivity are risks to check; do not record them as incidents without dated evidence.

Source and update notes

The original task list was attributed to the SAEON Trello board export supplied by Ethan Williams. This revision updates the documents and does not change the live Trello board. Archived cards remain excluded. Review the live board and evidence links before submission. Requirement reference: PRP37XS Subject Guide, pages 8 and 20.